

Radical-Induced In Situ Covalent Interfacial Grafting Coupled with Resin Infiltration Enhancement for High-Performance Thermoplastic Composites

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ABSTRACT

Poor interfacial bonding and insufficient resin infiltration currently represent two fundamental challenges that limit the structural performance of carbon fiber reinforced thermoplastic composites (CFRTPs). Herein, a radical induced interfacial reaction strategy is developed to simultaneously address these issues. Direct fluorination is employed to introduce thermally activatable radicals onto carbon fiber (CF) surfaces. During melt processing, these radicals preferentially induce scission at tertiary carbon sites along polypropylene (PP) chains, generating new radicals that subsequently couple with surface radicals on CF to form covalent interfacial bonds. Meanwhile, the reduction in molecular weight of interfacial PP lowers local melt viscosity, thereby improving resin infiltration into fiber tows. Importantly, this radical-induced activation is spatially confined to the interface, while the bulk PP matrix remains structurally unaffected. As a resulting composite exhibits

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significantly enhanced mechanical performance, including increases of 70% in tensile strength, 82% in flexural modulus, 50.1% in interlaminar shear strength, and 65.9% in fracture toughness compared with untreated CF/PP. In addition, porosity and water absorption decrease by 94.5% and 86.0%, respectively. This work establishes an effective strategy to overcome the long-standing limitations of weak interfacial bonding and poor resin infiltration in CFRTPs.

1. Introduction

Carbon fiber-reinforced polymer composites (CFRPs) have garnered widespread attention and application across various high performance sectors including rail transportation, aerospace, wind energy, and national defense owing to their lightweight nature, high specific strength and stiffness, as well as superior corrosion and thermal resistance[1, 2]. Depending on the type of polymer matrix, carbon fiber-reinforced polymer composites can be broadly divided into thermoset-matrix and thermoplastic-matrix systems. Thermoset-based composites initially dominated the market due to their excellent resistance to solvents and corrosive environments, along with superior thermal stability and mechanical strength [3–6]. However, with the growing demand for environmentally sustainable materials and the expanding application scope of composites, thermoplastic-based composites (CFRTPs) have emerged as a focal point of research and development[3-5]. These materials offer notable advantages, including superior impact resistance, enhanced damage tolerance, improved processability, design flexibility, and recyclability making them increasingly attractive for advanced engineering applications[6, 7].

Compared with conventional thermoset-based composites, the development of CFRTP is primarily constrained by two intrinsic challenges[8, 9]. First, their inherently high molecular

weight results in elevated melt viscosity, which hinders effective resin infiltration into fiber tows and promotes the formation of voids and other processing-induced defects[10-12]. Second, the low polarity of typical thermoplastic matrices leads to weak interfacial interactions with the reinforcement, resulting in poor interfacial compatibility[13-15]. Collectively, these limitations impede the overall performance enhancement of CFRTs, thereby restricting their broader application and further development.

To address the critical challenges of insufficient resin infiltration and weak interfacial bonding in CFRTs, various interfacial modification strategies have been developed. For example, surface modification of CF via plasma treatment or nanoparticle incorporation can effectively tailor surface morphology and roughness, thereby enhancing mechanical interlocking with the polymer matrix, improving surface energy, and ultimately promoting interfacial performance[15-18]. In addition, chemical grafting approaches that introduce specific functional groups onto CF surfaces can modulate surface chemical reactivity, facilitating the formation of stronger chemical interactions or covalent bonds at the fiber–matrix interface and substantially increasing interfacial bonding strength[19-23]. However, these approaches still have notable limitations. Plasma-based and nanoparticle-assisted treatments may introduce surface defects or structural damage to carbon fibers, potentially affecting their intrinsic mechanical properties[19-23]. Conventional chemical grafting mainly enhances interfacial adhesion through functional group introduction but does not resolve the high melt viscosity of thermoplastic resins, and thus provides limited improvement in resin infiltration into fiber tows[19-23]. In addition, some modification strategies involve bulk resin alteration, which may influence matrix properties and processing behavior[19-23]. and maintain the integrity of both the fibers and the polymer matrix.

Therefore, an alternative interfacial engineering strategy is needed that can simultaneously strengthen interfacial bonding, promote resin infiltration, and maintain the structural integrity of both the fibers and the polymer matrix.

Our previous studies demonstrated that direct fluorination can introduce radicals onto the surface of materials, and these radicals are thermally activated exhibit significant chemical reactivity[24]. In this work, direct fluorination was employed to modify CF, with the degree of fluorination precisely controlled through a regulated gas-phase fluorination process, thereby producing CF with radical-rich surfaces. PP was subsequently selected as the thermoplastic matrix for composite fabrication. PP offers several practical advantages, including low density, non-toxicity, excellent processability, low cost, and good chemical resistance[25-28]. More importantly, the tertiary carbon sites along PP molecular chains are highly susceptible to radical initiation, facilitating localized chain scission under thermal conditions. Following PP chain scission, the resulting radicals undergo coupling reactions with the surface-derived radicals on fluorinated CF, leading to the formation of covalent bonds at the fiber-matrix interface. Meanwhile, chain scission of PP in the interfacial region reduces the local molecular weight, thereby lowering melt viscosity near the fiber surface and facilitating resin infiltration into fiber tows. Importantly, this radical-induced activation is spatially confined to the interface, while the bulk PP matrix remains chemically intact, preserving its intrinsic structural and mechanical properties. Therefore, interfacial bonding strength and resin infiltration are simultaneously enhanced, resulting in significant improvements in the mechanical performance of the composites.

2. Experimental

2.1 Chemicals and Materials

Carbon fiber fabrics was obtained from Zhongfu Shenying Carbon Fiber Co., Ltd. (China). PP film (product designation: PP-H, GD, 040) was supplied by Maohua Shihua Co., Ltd. (China). Acetone were purchased from Chengdu KeLong Chemicals Co. Ltd. (China). The 20% F₂/N₂ mixture was obtained from Fujian Yongjing Technology Co., Ltd. (China). All chemicals were directly used without any purification.

2.2 Direct fluorination of carbon fibers

The direct fluorination process of Carbon fiber fabrics was shown in Fig. 1. The CF were initially washed with acetone using a Soxhlet extractor to remove the inherent sizing agents, and then dried at 120°C for 24 hours. The dried fabrics were transferred to a stainless-steel autoclave, where the atmosphere was replaced with pure N₂ three times to minimize contamination from residual gases. Subsequently, the autoclave was evacuated to create a vacuum. A gas mixture of F₂/N₂, with varying partial pressures, was introduced into the autoclave, where it reacted with the Carbon fiber fabrics for a specified duration. After the reaction, the gas mixture was evacuated and filtered through a tower. Finally fluorinated CF with different surface fluorine contents were obtained and the samples were labeled CF_{lf}, CF_{mf}, and CF_{hf} based on the fluoride concentration during the direct fluorination process, while the untreated fabrics were designated as CF.

2.3 Fabrication of carbon fiber reinforced composites

The preparation schematic of CF/PP composites was shown in Fig. 1. CF/PP composites were fabricated via a film-stacking and hot-pressing process. Untreated and fluorinated carbon fiber fabrics, including CF, CF_{lf}, CF_{mf}, and CF_{hf}, were first dried at 80 °C for 12 h to remove residual moisture prior to composite preparation. PP films were cut into sheets with

dimensions matching those of the carbon fiber fabrics. The composites were assembled by alternately stacking carbon fiber fabrics and PP films in a mold to achieve a designed fiber volume fraction of approximately 65%. Specifically, eight layers of carbon fiber fabrics and nine layers of PP films were sequentially stacked to ensure uniform resin distribution throughout the laminate. The stacked assemblies were then preheated in a hot press at 190 °C under a pressure of 3 MPa for 30 min to allow sufficient melting and infiltration of the PP matrix into the fiber tows. Subsequently, the temperature increased to 200 °C, and the pressure was raised to 5 MPa for 60 min to promote interfacial consolidation and composite densification. After hot pressing, the laminates were cooled to room temperature under a constant pressure of 5 MPa to minimize residual stresses and prevent void formation. The fully consolidated CF/PP composites were finally demolded and cut into specimens for subsequent characterization and mechanical testing.

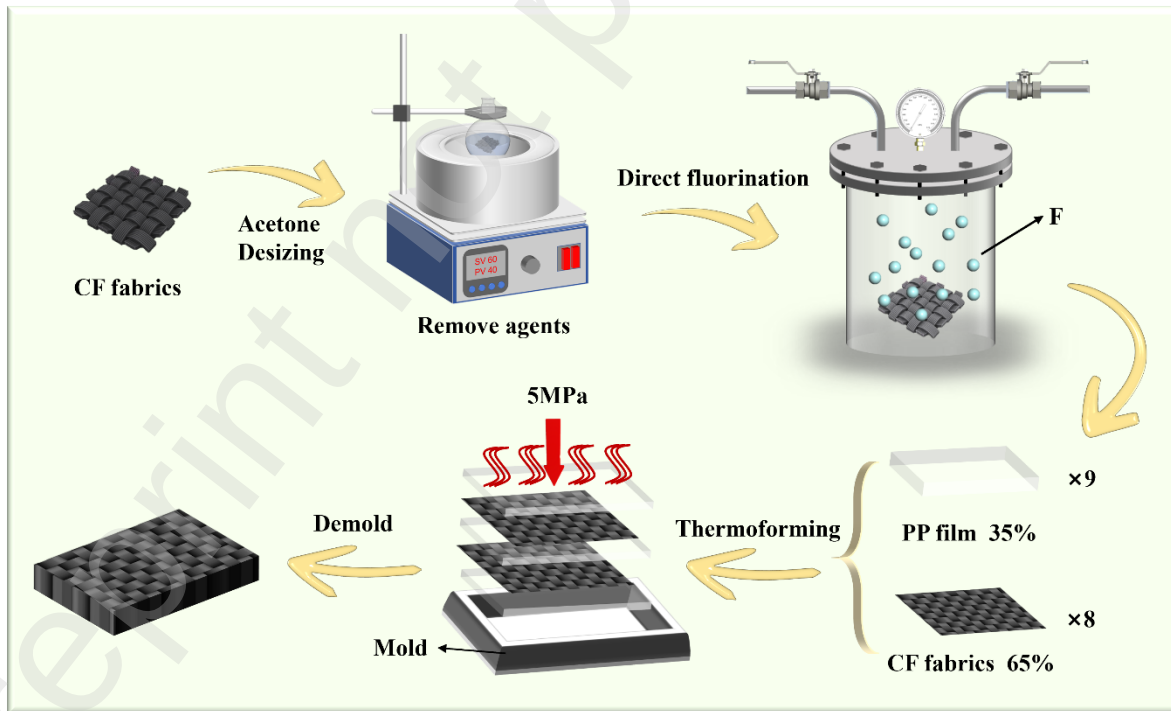


Fig. 1. Direct fluorination process of CF and the preparation schematic of CF/PP composites.

2.4 Characterization

Fiber surface chemical structure was analyzed using atomic force microscopy (AFM, Bruker, Icon, USA) can be used to further verify the surface structure of the samples. Surface elemental composition and detailed bonding information were further examined using X-ray Photoelectron Spectroscopy (XPS), which was performed on a Kratos XSAM800 spectrometer (Kratos Analytical Ltd., U.K.). Electron paramagnetic resonance (EPR, Bruker, EMXplus, Germen) was employed to confirm changes in radicals on the surface of fluorinated CF. To monitor the curing dynamics of fluorinated CF and PP resins, in situ Fourier transform infrared spectroscopy (FTIR) was employed to observe the changes in the chemical structure of the model compounds as the temperature increased, under N₂ protection. The in-situ FTIR spectrums were collected with the help of an external temperature controller (Yudian Co. Ltd., China), where the heating rate was set as 5°C/min and the spectra were collected by every 5 s in the region of 4000-400 cm⁻¹.

The scanning electron microscope (SEM, FEI, Apreo S HiVoc, EU/USA) were recorded the surface and cross-sectional structure of composites for directly observing the changes of morphology. The mechanical properties of single fiber and composites testing via KOL KDIII-5000 mechanical testing machine (Kaiqiangli, China). Each sample was measured ten times to get the average value. Dynamic mechanical analysis (DMA) was measured by three-point bending mode on TA Q800 with a heating rate of 10 °C/min. Solid-state NMR spectra were measured on Bruker AV III 500MHz spectrometer. Molecular weight of PP resin was tested by high temperature GPC system (Agilent, 1260 Infinity II, U.K.). Microscale X-ray computed tomography (Micro CT) was performed by vivaCT80 (Scanco, Switzerland).

3. Results and Discussion

3.1 Structure and Performance Characterization of fluorinated CF

The morphological characteristics of carbon fibers subjected to varying degrees of fluorination were analyzed using SEM, as shown in Fig. 2(a1-a4). The results indicate that fluorine introduction does not cause noticeable damage to the fiber surface, which remains relatively smooth both before and after fluorination. To further evaluate surface roughness, AFM measurements were performed (Fig. 2(b1-b4)). The AFM images reveal the absence of prominent surface grooves across all fiber types, a feature typical of high-strength carbon fibers produced via the wet-dry-spraying method. Quantitative analysis of surface roughness (Ra) showed minimal variation, with values ranging from 2.81 to 2.07 nm before and after fluorination. These findings confirm that the direct fluorination process does not significantly alter the surface morphology of carbon fibers.

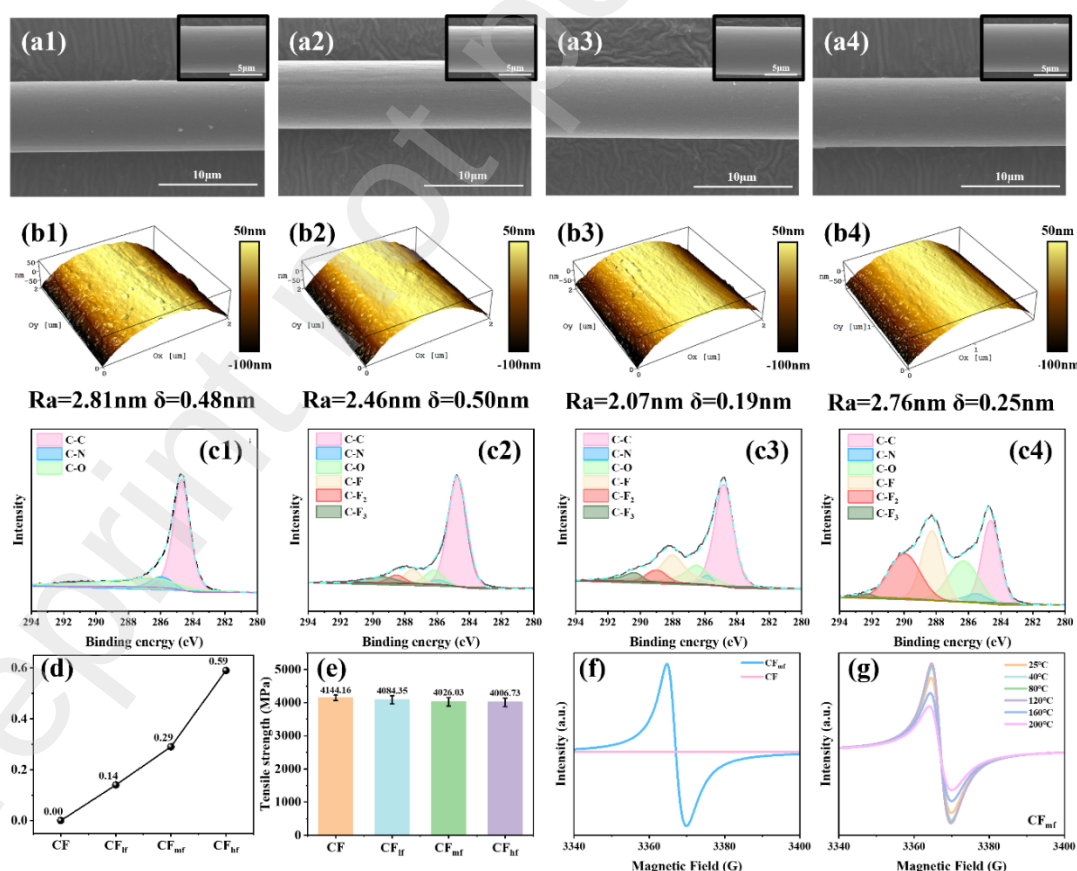


Fig. 2. (a1-a4) SEM images (b1-b4) AFM images and roughness analysis (c1-c4) XPS C1s peak spectra. (d) Fluorine to carbon atom ratio (e) Tensile strength (f) EPR spectrums of CF and CF_{mf}. (g) In-situ EPR spectrums of CF_{mf}.

X-ray photoelectron spectroscopy (XPS) analysis of the unfluorinated carbon fiber (CF) revealed three signal peaks in the C 1s spectrum (Fig. 2(c1)), corresponding to C=C/C-C (≈ 284.8 eV), C-N (≈ 285.9 eV), and C-O (≈ 286.4 eV). These results suggest the presence of trace N on the fiber surface, which is analyzed as the sizing agent was not removed cleanly resulting in some sizing agent residues. Following direct fluorination, three additional peaks appeared in the C 1s spectrum at C-F (≈ 287.5 eV), C-F₂ (≈ 288.6 eV), and C-F₃ (≈ 292.2 eV), which are attributed to various C-F bonding configurations (Fig. 2(c2-c4)). This confirms the successful incorporation of fluorine atoms onto the CF surface, which was equally illustrated by the results of EDS and FTIR (Fig. S1, S2). Furthermore, wide-scan XPS spectra (Fig. S1) demonstrated a progressive increase in fluorine content with higher degrees of fluorination. The ratio of fluorine to carbon atomic for the low, medium, and high fluorinated CF samples (CF_{lf}, CF_{mf}, and CF_{hf}) were 0.14, 0.29, and 0.59, respectively (Fig. 2 (d)).

Surface modification typically has a significant impact on the tensile properties of fibers. In our previous study, direct fluorination was identified as a promising non-destructive surface treatment strategy[29]. As shown in Fig. 2(e), the tensile test results of the fluorinated CF demonstrated stable mechanical strength with increasing surface fluorine content, thereby ensuring the excellent mechanical performance of the resulting reinforced composites. This observation is further corroborated by Raman spectroscopy results (Fig. S3), where the redshift of the G peak and the I_D/I_G ratio (from 1.22 to 1.87) confirm the successful incorporation of fluorine and indicate that direct fluorination does not damage the carbon

fiber structure. All these results demonstrate the controllability of the direct fluorination method, allowing the fluorination process to occur only on the surface of the CF.

The chemical environment evolution of CF surface before and after fluorination at elevated temperatures was investigated using EPR spectroscopy. No detectable radical signal is observed in pristine CF, whereas CF_{mf} exhibits a pronounced signal with a g-factor around 2.0, characteristic of carbon-centered radicals (Fig. 2(f)). Upon heating, the radical signal intensity in CF_{mf} initially increased and subsequently decreased (Fig. 2(g)). This is because at the initial stage of warming, the increase in temperature provides the energy required for radical expansion, leading to the propagation reactions of radical. After ramping up to 120 °C, the random combination of the two radical centers leading to a decrease in the intensity of the radical signals. The free radicals present on the surface of fluorinated CF provide a favorable basis for subsequent covalent grafting at the fiber–matrix interface.

3.2 Radical-induced interfacial grafting and reaction mechanism

EPR analysis was employed to monitor the evolution of free radicals during the compounding of fluorinated CF with PP. The CF_{mf}/PP sample, representing an intermediate fluorination level, was selected for detailed investigation. As shown in Fig. 3(a), with increasing temperature, the free-radical signal intensity of the CF_{mf}/PP samples initially increased and subsequently decreased, indicating a sequential process of radical activation followed by quenching during melt compounding. This temperature-dependent radical evolution creating favorable conditions for the formation of covalent bonds at the fiber–matrix interface. To elucidate the origin of the pronounced free radical quenching effect, in-situ heating FTIR spectrums analyses were conducted on CF_{mf}/PP, to simulate the elemental and structural changes occurring within the composites during the molding process (Fig. 3(c-

e)). The results indicate that sample exhibit fluctuations in absorption peaks within the 1100-1200 cm^{-1} range. It was observed that the C-C stretching vibrational absorption peaks of $\text{CF}_{\text{mf}}/\text{PP}$ underwent shifts during the heating process. Noda's rules effectively clarify the fluctuations in peak intensity that occur within both synchronous (left column) and asynchronous (right column) correlation spectra of C-C, with red and blue regions representing positive and negative correlation intensities, respectively[30]. For $\text{CF}_{\text{mf}}/\text{PP}$, where $\Phi(1166.4, 1146.3) > 0$ and $\Psi(1166.4, 1146.3) > 0$, it suggests that the peak gradually shifts from 1166.4 cm^{-1} to 1146.3 cm^{-1} during heating, indicating changes in their chemical environments. Considering the earlier observations of radical evolution in $\text{CF}_{\text{mf}}/\text{PP}$ during heating, it is inferred that radicals generated from CF_{mf} initiated C-C chain scission in PP, resulting associated shifts in absorption peaks.

The use of ^{13}C NMR spectroscopy further confirmed that a free radical-induced chain-breaking and coupling process occurred during the complexation of fluorinated CF with PP. ^{13}C NMR spectra were obtained for both pure PP and "Break PP" (extracted used xylene on $\text{CF}_{\text{mf}}/\text{PP}$ at 130 $^{\circ}\text{C}$). As shown in Fig. 3(b), three characteristic peaks were observed at 21.46, 25.92, and 43.78 ppm, corresponding to $-\text{CH}_2-$, $-\text{CH}-$, and $-\text{CH}_3-$ groups, respectively. In pure PP, the relative intensities of these peaks were 27% ($-\text{CH}_2-$), 50% ($-\text{CH}-$), and 23% ($-\text{CH}_3-$). In Break PP, the respective intensities shifted to 28%, 47%, and 25%. The decrease in $-\text{CH}-$ content and the increase in $-\text{CH}_2-$ and $-\text{CH}_3-$ content in Break PP compared to pure PP further demonstrated that the addition of fluorinated CF triggered the PP chain breakage to occur at the tertiary carbon sites in the PP molecular chain.

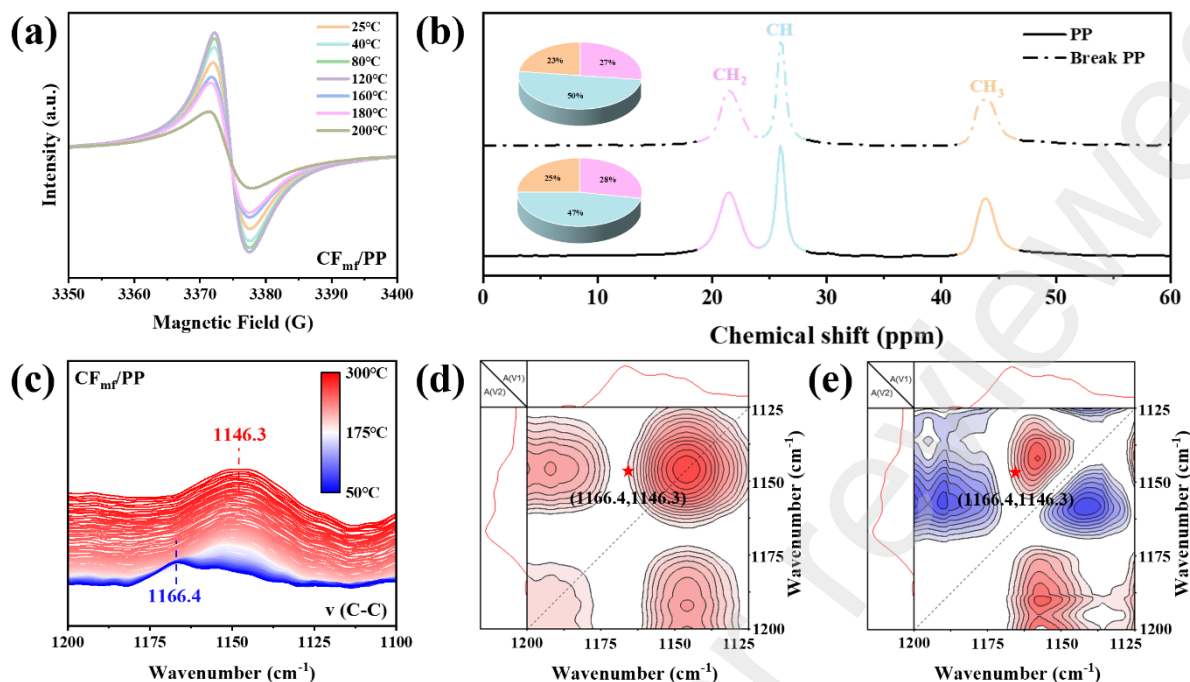


Fig. 3. (a) In-situ EPR spectrums of $\text{CF}_{\text{mf}}/\text{PP}$. (b) ^{13}C NMR spectra of PP and Break PP. (c) The dynamic infrared spectrum and peak shift of $\text{CF}_{\text{mf}}/\text{PP}$. (d, e) The synchronous and asynchronous infrared spectra of $\text{CF}_{\text{mf}}/\text{PP}$.

The composite samples CF/PP , $\text{CF}_{\text{lf}}/\text{PP}$, $\text{CF}_{\text{mf}}/\text{PP}$, and $\text{CF}_{\text{hf}}/\text{PP}$ were repeatedly washed with xylene to remove PP, yielding fiber products denoted as w-CF, w- CF_{lf} , w- CF_{mf} , and w- CF_{hf} , respectively. Surface morphology observations of w-CF revealed a clean fiber surface with no detectable PP residue. In contrast, residual PP was clearly observed on the surfaces of w- CF_{lf} , w- CF_{mf} , and w- CF_{hf} . Notably, PP remaining on w- CF_{lf} was observed as discrete particles sparsely distributed on the fiber surface, indicating limited residual polymer after solvent washing and thus a relatively low grafting density. In contrast, w- CF_{mf} and w- CF_{hf} exhibited a more continuous PP layer firmly adhered to the fiber surfaces, reflecting a higher amount of retained polymer and suggesting increased grafting density (Fig. 4(a1-a4)). These observations indicate that direct fluorination promotes interfacial reactions between CF and PP, leading to enhanced interfacial bonding. Moreover, the extent of PP grafting shows a

positive correlation with fluorination level. The degree of grafting between CF and PP was quantitatively assessed using a gravimetric method, and the corresponding grafting rates were calculated. The results showed that the grafting rates for CF/PP, CF_{lf}/PP, CF_{mf}/PP, and CF_{hf}/PP were 0.2%, 6.6%, 9.4%, and 8.7%, respectively (Fig. (4b)). These findings indicate that the fluorination treatment substantially enhanced the grafting efficiency of CF to PP. However, excessive fluorination, as in the case of CF_{hf}, resulted in a slight reduction in grafting rate, suggesting that over-fluorination may hinder the formation of effective interfacial bonding between CF and PP.

IR spectral analysis, as shown in Fig. 4(c), revealed characteristic absorption peaks in CF corresponding to -CH₂ groups at 3000.1 cm⁻¹ and -CH₃ groups at 2890.7 cm⁻¹, originating from the intrinsic backbone structure of the carbon fibers. In contrast, for w-CF and w-CF_{mf}, only -CH₃ absorption bands were observed within the ranges of 2836.5-2838.7 cm⁻¹ and 1374.9-1450.2 cm⁻¹, which are attributed to PP molecular chains grafted onto the fiber surface. Notably, the intensity of the -CH₃ absorption peaks increased with the degree of fluorination, indicating a progressive enhancement in PP grafting. These observations confirm that a grafting reaction occurred at the interface between the fluorinated carbon fibers and PP resin during the hot-pressing process, resulting in the formation of covalent bonds that improve interfacial adhesion.

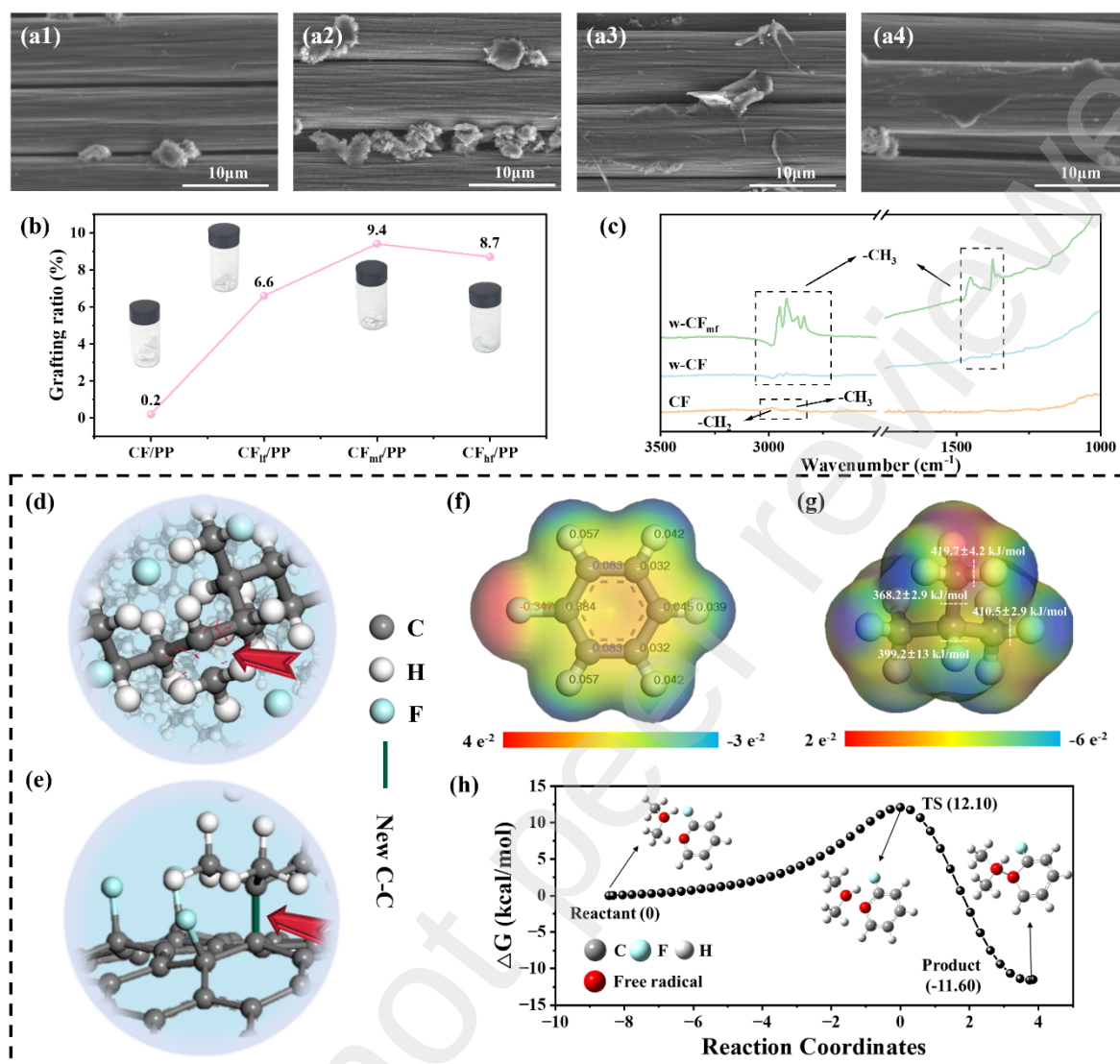


Fig. 4. (a1-a4) SEM images of w-CF, w-CF_{lf}, w-CF_{mf}, w-CF_{hf} (fibers in CF, CF_{lf}, CF_{mf}, CF_{hf} washed by xylene). (b) Grafting ratio of CF, CF_{lf}, CF_{mf}, CF_{hf}. (c) IR spectrums of CF, w-CF, w-CF_{mf}. Insets show the corresponding Break PP power. (d) Schematic diagram of the radical induce chain scission mechanism. (e) Schematic diagram of the radical induced graft covalent bonding. (f) Electrostatic of fluorobenzene molecule. (g) Electrostatic and bond energy of PP molecules. (h) Reaction transition state and path energy diagram of carbon free radical in fluorobenzene and PP.

Based on the above analyses, it can be concluded that the composite formation between fluorinated CF and PP involves two distinct stages: chain scission (Fig. 4(d)) and grafting (Fig. 4(e)). After fluorination of the carbon fiber surface, the strong electron-withdrawing

nature of fluorine induces significant polarization of the adjacent C–C bonds, thereby increasing the chemical reactivity of the neighboring carbon atoms. This electronic effect promotes the generation of radical-active sites on the fiber surface, which are more prone to participate in subsequent reactions. The simulated molecular structure shown in Fig. 4(f) provides further support for this mechanism by revealing the altered electronic environment around the fluorinated carbon sites. The chain scission of PP is initiated by free radicals present on the fluorinated CF surface, which preferentially attack tertiary carbon atoms in the PP backbone (Fig. 4(g)), leading to bond cleavage and the formation of new radical species[31]. These newly generated radicals then rapidly couple with the original radicals on the carbon fiber surface, resulting in the formation of covalent bonds. Fig. 4 (h) presents the simulated chemical reaction pathway of model molecules containing free radical species. The corresponding results supporting the mechanism discussed above.

3.3 Radical-Induced Enhancement of Interfacial Resin Infiltration

We have demonstrated that fluorination-induced radicals can initiate in situ covalent grafting at the CF/PP interface. Beyond interfacial chemical bonding, an improvement in resin infiltration was also observed. The infiltration behavior of PP resin into CF fabrics with different fluorination levels is shown in Fig. 5(a1–a4). A 7-mm-thick PP block was placed on CF fabrics and heated at 180 °C for 30 min, after which the penetration of molten PP into the fiber tows was evaluated. The thickness of the residual PP block after infiltration (denoted as H) was used as a quantitative indicator, where a smaller H value corresponds to greater resin infiltration. Under identical conditions, fluorinated samples exhibited consistently lower H values than the unmodified sample, indicating enhanced resin infiltration. Among them, the CF_{mf}/PP sample showed the lowest H value (5.50 mm), corresponding to an 11.6%

reduction compared with CF/PP (6.22 mm), demonstrating a pronounced improvement in permeability.

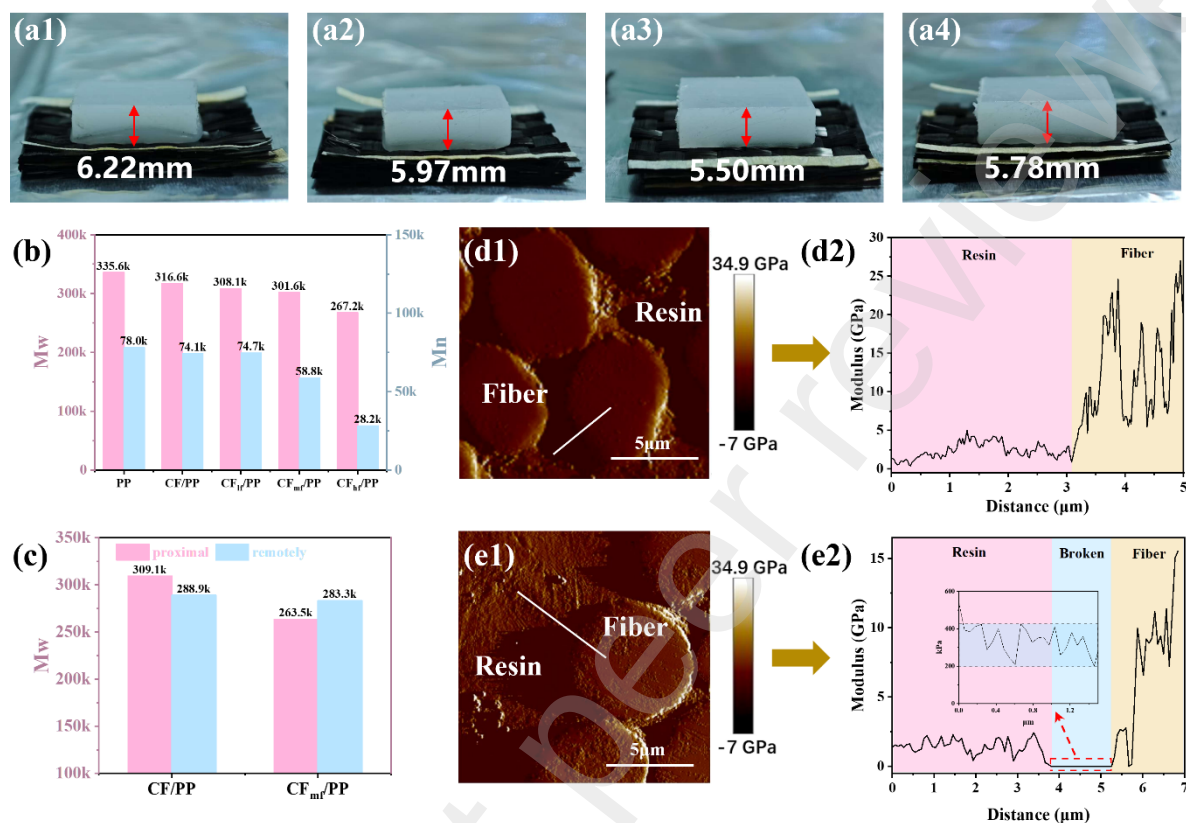


Fig. 5. (a1-a4) Optical images showing the resin infiltration behavior of PP with CF, CF_{lf}, CF_{mf}, and CF_{hf}. (b) Mw and Mn of PP in composites. (c) Remotely and proximal Mw of CF/PP and CF_{mf}/PP. (d1, d2) The modulus mapping and modulus transition curves of CF/PP. (e1, e2) The modulus mapping and modulus transition curves of CF_{mf}/PP.

The GPC results offer insight into the behavior observed in Fig. 5(a). As shown in Fig. 5(b), the weight-average molecular weight (Mw) and number-average molecular weight (Mn) of pure PP are approximately 335.6 kDa and 78 kDa, respectively. When PP was composited with fluorinated CF, both Mw and Mn decreased, and the reduction in molecular weight became significantly more pronounced with increasing degrees of fluorination. To further explore the interfacial infiltration enhancement effect of fluorinated CF/PP, PP resin

films of two different thicknesses (thin and thick) were composited with single-layer CF fabrics and subjected to high-temperature GPC. Molecular weights were compared between the PP regions near the fiber (proximal, $<30\ \mu\text{m}$ from the fiber surface) and those far from the fiber (distal, $>300\ \mu\text{m}$ from the fiber surface), as shown in Fig. 5(c). For the CF/PP composite, the molecular weight at the distal end was lower than that at the proximal end. This is attributed to oxidative degradation at the distal region, which directly contacted the heated platen during thermoforming. However, in the fluorinated CF/PP composites, the molecular weight of proximal PP was significantly lower than that of the distal region. This reversal confirms that radical-induced chain scission is localized at the fiber–matrix interface during molding, without significantly altering the bulk resin. Such interfacial molecular weight reduction lowers the local melt viscosity, thereby facilitating resin infiltration into fiber tows.

To more directly visualize resin infiltration at the fiber–matrix interface, AFM was employed to probe the interfacial modulus distribution between CF and PP, as shown in Fig. 5(a, b). In the modulus maps of the unfluorinated CF/PP composite (Fig. 5(a1, a2)), a sharp modulus contrast is observed at the fiber–resin interface, with an abrupt transition and virtually no interfacial gradient zone. This indicates a lack of mechanical continuity and weak interfacial integration. In contrast, the CF_{mf}/PP composite exhibits an intermediate, ultra-low modulus region between the fiber and resin phases, with a width of approximately $1\ \mu\text{m}$ (Fig. 5(b1, b2)). As the variation in the resin modulus falls outside the measurable range of the probe, the quantified values cannot reliably reflect the true modulus. Nevertheless, the presence of a distinct low-modulus region is clearly confirmed. This transitional region is consistent with the previously described zone of PP chain scission induced by radicals on the

fluorinated CF surface. Overall, these observations provide further support for the existence of a chemically modified interfacial region that facilitates both covalent bonding and enhanced resin infiltration.

Improved resin infiltration can significantly influence the evolution of internal pore structures. Accordingly, the pore morphology and distribution within the composites were systematically investigated, and the results are presented in Fig. 6(a1–a4). In the images, the shiny filamentous regions represent the weft fibers, the rounded regions correspond to the radial fibers, the grey areas indicate the PP resin matrix, and the black regions denote pores. To quantitatively assess pore distribution, the Pore (Particle) Image Recognition and Analysis System (PCAS) was employed for cross-sectional image analysis[32]. In the processed binary images shown in Fig. 6(b1–b4), the white regions represent pores, while the black regions correspond to the bulk composite material. The CF/PP composite exhibits several large pores, with individual diameters exceeding 50 μm . Following fluorination treatment, the pore population is markedly reduced, indicating a significant densification of the composite structure. However, in the CF_{lf}/PP composite, pore formation reemerges. Four laminated board specimens (5 mm $\times$ 5 mm) were subjected to three-dimensional scanning. The Micro-CT results presented in Fig. 6(c1–c4) clearly reveal the same trend, providing three-dimensional structural evidence for the pore evolution behavior. As shown in Fig. 6(d), the porosity of CF/PP is 1.09%. In the CF_{lf}/PP sample, the overall porosity decreases to 0.64%. A further reduction is observed in CF_{mf}/PP, which porosity as low as 0.06%. The water absorption behavior of the composites exhibits a similar trend (Fig. 6(e)). These observations indicate that radical-induced interfacial reactions promote resin infiltration and suppress pore formation, resulting in progressive structural densification (Fig. 6(f)).

Nevertheless, excessive fluorination may lead to over-encapsulation of the fiber surface by fluorine-containing groups, which impedes effective interfacial bonding with the resin and consequently increases porosity.

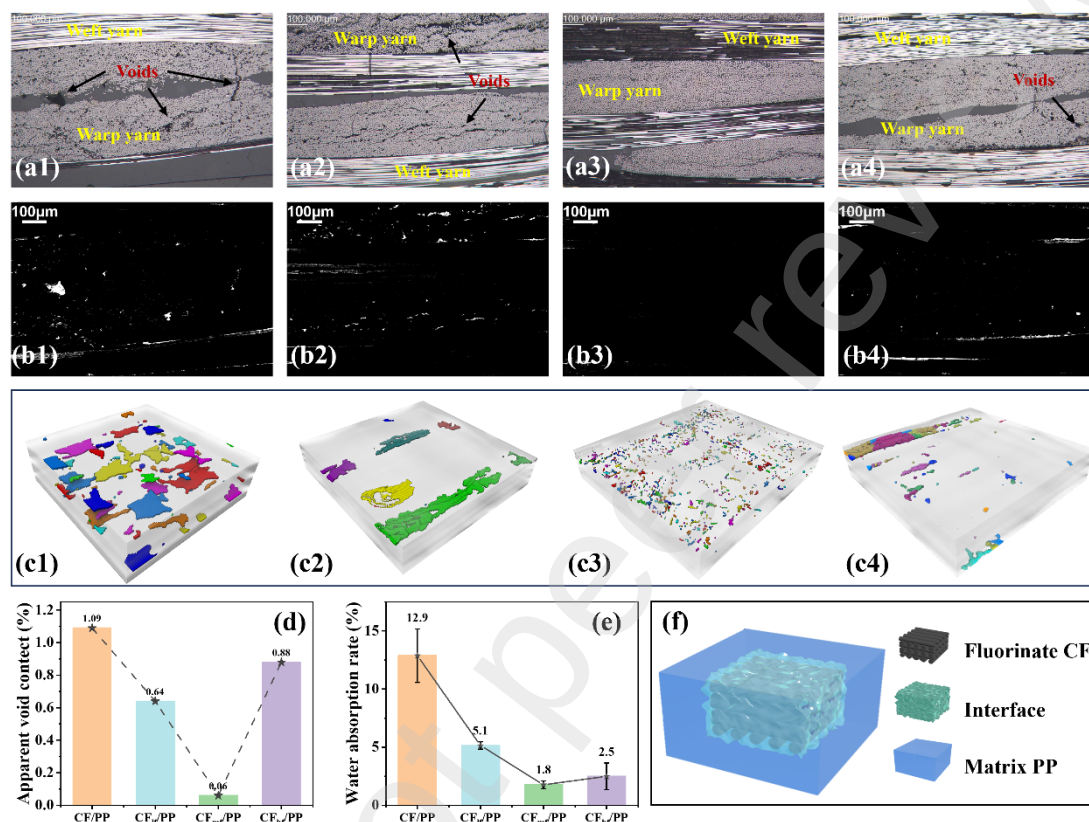


Fig. 6. (a1-a4) Cross-sectional optical images, (b1-b4) Void image calculated using PCAS, (c1-c4) Three-dimensional reconstruction results from Micro CT measurement. (d) Apparent void content and (e) Water absorption rate of CF/PP, CF_{lf}/PP, CF_{mf}/PP, and CF_{hf}/PP. (f) Schematic diagram of the interface between fluorinated carbon fiber and PP resin matrix.

3.4 Mechanical performance of CF/PP composites

Mechanical properties were fundamental indicators for evaluating the application potential of composite materials. To investigate the influence of direct fluorination on the mechanical performance of CF/PP composites, both tensile and flexural tests were conducted.

The results demonstrated that the incorporation of fluorinated carbon fibers significantly enhanced the mechanical properties of the composites. Among the samples tested, CF_{mf}/PP exhibited the most notable performance improvements. The tensile strength increased from 174.58 MPa (CF/PP) to 296.33 MPa, representing a 70% enhancement. Likewise, the tensile modulus increased from 5.05 GPa to 7.12 GPa, corresponding to a 41% improvement (Fig. 7a). These enhancements are primarily attributed to the strengthened interfacial bonding between the fluorinated fibers and the PP matrix, which facilitates efficient load transfer. A similar trend was observed in the flexural tests. CF_{mf}/PP displayed a maximum flexural strength of 160.26 MPa, compared to 118.40 MPa in the unmodified CF/PP, reflecting a remarkable 35.4% increase. The flexural modulus also increased substantially, from 17.96 GPa to 32.69 GPa—an enhancement of 82.0% (Fig. 7b). These substantial improvements in bending properties are attributed to the reduction of internal porosity and the overall densification of the composite structure, both of which are facilitated by improved interfacial compatibility. The ILSS and impact resistance tests show the same trend, with improvements of 50.1% and 29.0% respectively (Fig. 7c, d).

Toughness is also a critical performance metric for carbon fiber composites in practical applications. To evaluate this property, fracture toughness tests were conducted using the critical stress intensity factor (K_{IC}) as the assessment criterion. The K_{IC} value of CF_{mf}/PP increased by 65.9%, rising from approximately 2.14 MPa·m^{1/2} in CF/PP to approximately 3.55 MPa·m^{1/2}, as shown in Fig. 7(e). This significant enhancement in fracture toughness is primarily attributed to the improved interfacial adhesion between the fluorinated carbon fibers and the PP matrix. Fig. 7(f) shows a comparison of the mechanical properties of all the above CF/PP and CF_{mf}/PP.

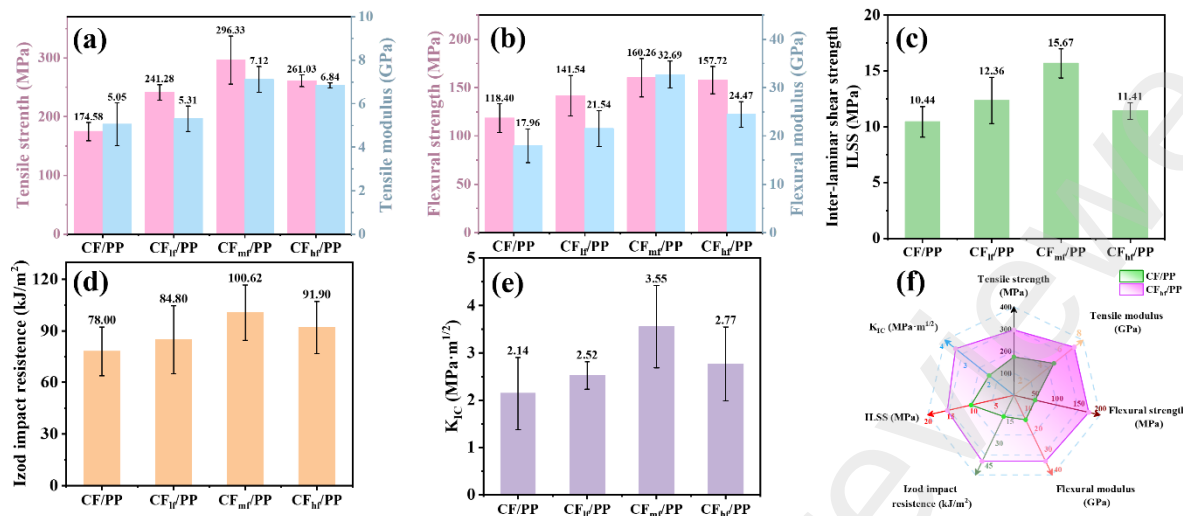


Fig. 7. Mechanical properties of CF/PP composite of (a) Tensile properties (b) Flexural properties (c) ILSS (d) Impact strength and (e) K_{IC} value. (f) Comparison of mechanical properties of CF/PP and CF_{mf}/PP.

SEM was employed to observe the latitudinal and radial fracture surfaces of the composites, as shown in Fig. 8. Examination of the radial fracture surface of the CF/PP composite (Fig. 8(a1)) reveals numerous voids and pull-out cavities left by the detached carbon fibers, along with evident interfacial gaps. The latitudinal fracture surface of the CF/PP composites (Fig. 8(a2)) exposes a very smooth fiber surface, and the resin and CFs are completely stripped away along the interface. These features indicate poor interfacial adhesion, suggesting that the bonding between the unfluorinated CF and the PP resin is weak and insufficient to resist fiber pull-out under mechanical stress (Fig. 8(a3)). Following fluorination of the carbon fibers and their incorporation into the PP matrix, the radial fracture surfaces of the resulting composites (Fig. 8(b1, c1)) exhibited relatively flat morphologies with significantly reduced fiber pull-out and interfacial debonding. Notably, no discernible gaps were observed between the CF and the PP resin, indicating improved interfacial stress transfer and more uniform load distribution across the fibers. On the latitudinal fracture

surfaces (Fig. 8(b2, c2)), a substantial amount of resin was found adhering to the fiber surfaces, characterized by fish scale-like cracking patterns. The carbon fibers were entirely embedded in resin fragments, suggesting that, during mechanical failure, crack propagation occurred predominantly within the resin matrix rather than along the fiber-matrix interface (Fig. 8(b3, c3)). This observation further confirms the enhancement of interfacial bonding and energy dissipation mechanisms as a result of fluorination treatment. However, in the case of over-fluorination, the excessive presence of fluorine-containing functional groups on the CF surface negatively impacts the interfacial properties of the composite. As evident in the radial (Fig. 8(d1)) and latitudinal (Fig. 8(d2)) fracture surfaces of the CF_{hf}/PP composite, visibly fractured carbon fiber fragments can be observed. This phenomenon is attributed to the overabundance of fluorine-containing groups encapsulating the fiber surface, which inhibits effective interfacial bonding with the PP resin. Consequently, stress transfer across the interface is significantly impaired, leading to premature fiber failure under mechanical loading (Fig. 8(d3)).

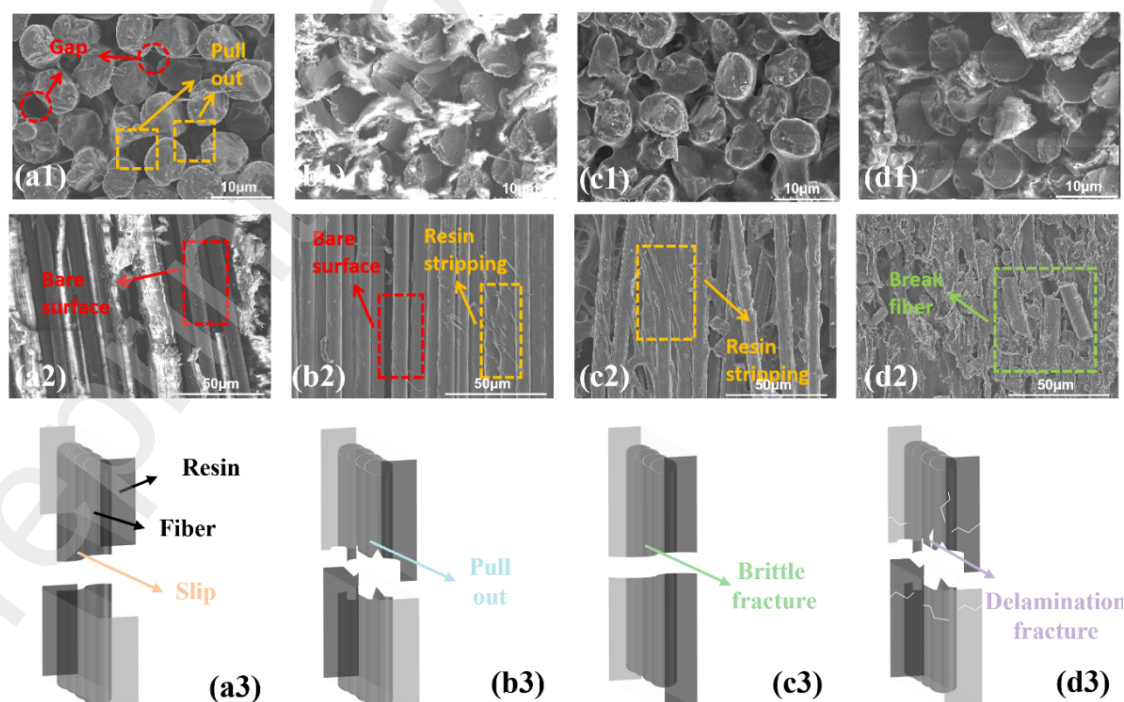


Fig. 8. SEM images of the radial and latitudinal fracture surfaces and schema of (a1-a3) CF/PP, (b1-b3) CF_{lf}/PP, (c1-c3) CF_{mf}/PP, (d1-d3) CF_{hf}/PP.

DMA test was conducted to evaluate the interfacial behavior of CF/PP composites. As shown in the $\tan\delta$ curve in Fig. 9(a), four characteristic transitions were observed, including the glass transition temperature (T_g), amorphous region relaxation (α_a), crystalline region relaxation (α_c), and melting temperature (T_m). In the CF_{lf}/PP composite (Fig. 9(b)), the same four relaxation peaks were present, but their intensities were noticeably reduced, indicating improved compatibility between the carbon fibers and the PP matrix, and more efficient interfacial stress transfer. This trend became more pronounced in CF_{mf}/PP and CF_{hf}/PP composites (Figs. 9(c, d)), where the T_g , α_a , and α_c relaxation peaks nearly disappeared. This attenuation of relaxation signals suggests a significant enhancement in interfacial bonding, leading to restricted polymer chain mobility near the fiber surface. Additionally, compared to CF/PP, both T_g and T_m were found to decrease in CF_{lf}/PP and CF_{hf}/PP, whereas CF_{mf}/PP exhibited an increase in these transition temperatures. The reduction in T_g and T_m is attributed to premature phase transitions and melting caused by PP chain scission. However, in CF_{mf}/PP, the enhanced interfacial interactions effectively suppressed the detrimental effects of chain breakage, resulting T_m increase. Furthermore, the initial storage modulus, which reflects strength of bonding at the interface, increased substantially upon fluorination. While the CF_{mf}/PP demonstrated the highest initial storage modulus, consistent with the strongest interfacial bonding observed in previous characterizations. In summary, the composite reinforcement mechanism involves both strong interfacial bonding and compliant interfacial regions. Covalent bonds formed at the fiber–resin interface promote effective stress transfer to the carbon fibers under impact or shear loading, while the low-modulus interfacial layer

contributes to energy dissipation and deformation accommodation, thereby mitigating interlaminar crack propagation (Figs. 9(e)).

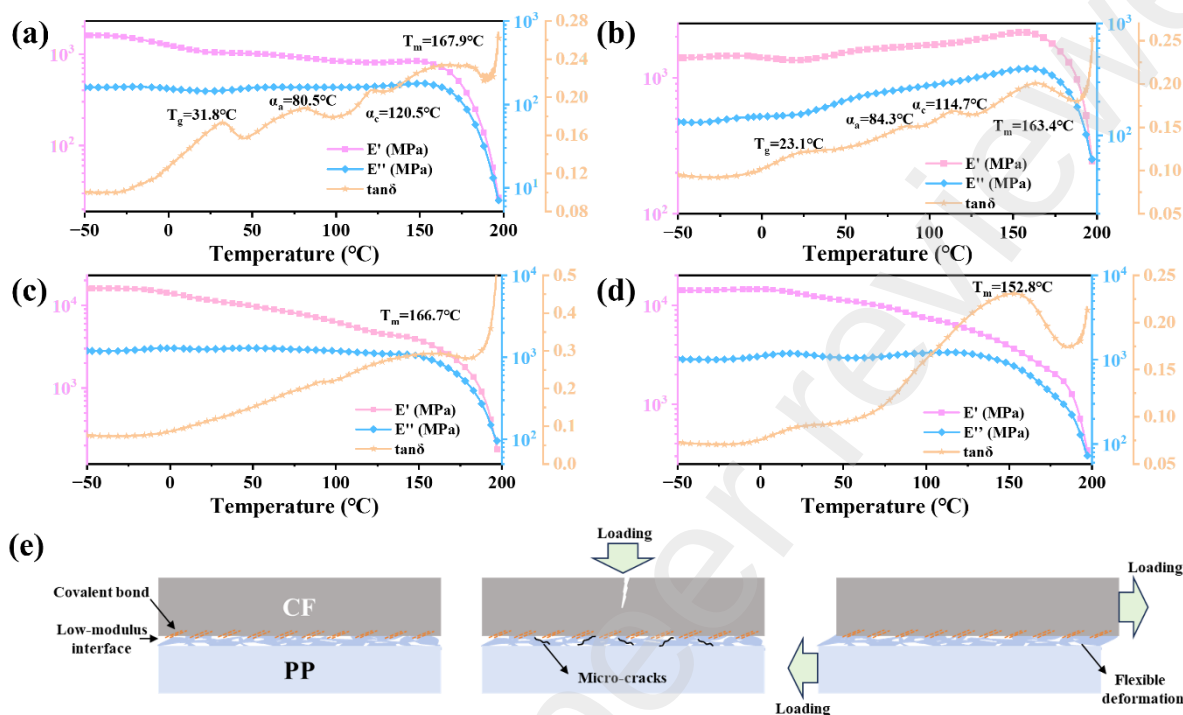


Fig. 9. (a-d) Storage modulus, loss modulus and $\tan\delta$ of CF/PP, CF_{lf}/PP, CF_{mf}/PP, CF_{hf}/PP. (e) Schematic representation of the mechanism for enhancing the interface in CF_{mf}/PP composites.

4. Conclusions

In summary, in this study, carbon fibers were successfully modified using a controlled direct fluorination technique and employed to enhance the mechanical performance of PP resin composites. As a result of this interfacial engineering approach, the composites exhibited simultaneous improvements in key mechanical properties, including tensile strength (296.33 MPa), tensile modulus (7.12 GPa), flexural strength (160.26 MPa), flexural modulus (32.69 GPa), ILSS (15.67 MPa), impact strength (100.62 kJ/m²), and fracture toughness K_{IC} (3.55 MPa·m^{1/2}). These enhancements arise from radicals generated on the fluorinated carbon fiber surface, which induce localized PP chain scission at the interface.

The resulting radical sites enable covalent fiber–matrix bonding, while the interfacial molecular weight reduction lowers local melt viscosity and promotes resin infiltration. Notably, this activation is confined to the interface, leaving the bulk PP matrix structurally intact. Together, strengthened interfacial bonding and improved infiltration lead to enhanced mechanical performance. This work presents an interfacial engineering strategy that simultaneously improves bonding and infiltration without compromising matrix integrity, offering a practical pathway for high-performance thermoplastic composites.

Notes

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